The Panorama of Acetylene Production and Its Potential Renaissance as a Building Block

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Abstract

Acetylene, produced from coal or natural gas, was a cornerstone of the chemical industry until the 1960s due to its rich triple bond chemistry. However, the long-term global availability of inexpensive petroleum before 1973 accelerated the production of olefins, such as ethylene, diminishing acetylene's market prominence due to their similar applications in downstream chemical production. As petroleum prices fluctuate, acetylene has regained economic viability in certain regions, notably accounting for approximately one-third of the global vinyl chloride production, despite its high carbon footprint. Environmental and economic assessment shows that replacing coal with biochar in acetylene-derived vinyl chloride production not only lowers carbon footprint but could also remain economically viable compared to the traditional ethylene route. Despite this potential, research on acetylene has been largely overlooked for decades. This review provides a holistic overview of acetylene production technologies, proposes sustainable production initiatives and quantifies their economic and environmental performance, and explores potential applications. By showcasing this promising trajectory, we seek to rekindle interest and foster collaborative efforts in advancing sustainable acetylene production and broadening its use as a feedstock for bulk chemicals.

Keywords

Acetylene; Catalyst design; Reactor design; Renewable carbon sources; Environmental and economic assessment.

1. Introduction

Acetylene was accidentally discovered in 1836 by Edmund Davy in Ireland through the reaction of water with potassium carbide (**Fig. 1**).[1] Significant study only began after its rediscovery by the French chemist Marcellin Berthelot who coined the name “acetylene” in 1860 and reported its synthesis at high temperatures using an electric arc in 1862.[2] The first commercial calcium carbide (CaC_2) plant for acetylene (C_2H_2) production was established in 1894, following the discovery by Thomas Willson of CaC_2 production through the electrothermal reduction of lime (CaO) with carbon sources and its simple reaction with water to form C_2H_2.[3] Early C_2H_2 applications were mainly in lighting and oxyacetylene welding and cutting due to the brightness and high temperature of its flame.[2] The importance of C_2H_2 in the chemical industry stems from its triple bond, which makes the molecule reactive and offers rich chemical possibilities. Industrial acetaldehyde production *via* C_2H_2 hydration in 1916, along with studies on C_2H_2 chemistry by Walter Reppe to produce various commodity chemicals, positioned C_2H_2 as a pillar of the chemical industry.[4] Coal-based CaC_2 was the sole source for C_2H_2 production until the development of commercial light hydrocarbon cracking methods, including the electric arc (Hüls arc) process in 1940, and the partial combustion (PC) and Wulff thermal pyrolysis (TP) processes in 1950.[5,6]

The birth and rapid growth of the petrochemical industry in the USA during the 1940s and in Europe during the 1950s, coupled with the safer handling of olefins compared to C_2H_2, led to the development of more cost-effective steam cracking technologies for olefin production, gradually displacing the C_2H_2 market primarily with ethylene (C_2H_4).[7,8] Quantitatively, global production of C_2H_2 and C_2H_4 reached 3.0 and 3.5 million metric tons per year (Mt y^{-1}) by 1960, respectively,[9–11] and these figures rose to 11 Mt y^{-1} for C_2H_2 and a staggering 225 Mt y^{-1} for C_2H_4 by 2021.[12–14] Between the 1970s and 1980s, PC became the dominant technology for global C_2H_2 production, as other methods were mostly abandoned. In recent years, China has seen a resurgence in CaC_2 production, rising from ca. 3 Mt y^{-1} in 1998 to 28 Mt y^{-1} in 2020.[15] This growth is linked to a significant increase in polyvinyl chloride (PVC) production, from less than 1.6 Mt y^{-1} in 1998 to over 20 Mt y^{-1} in 2020.[16,17] China now leads global C_2H_2 production (>8 Mt y^{-1}), accounting for over 70% of world production.[13,14,18] The historical transition from acetylene to ethylene derived from the long-term availability of inexpensive oil before 1973 (**Fig. 1**).[19,20] As oil prices fluctuate, C_2H_2 production has become economically viable in regions where oil prices are high and electricity, coal, or natural gas costs are low,[21] leading to the recent increase in C_2H_2 production in China. Additionally, the triple bond of C_2H_2 makes it unique for certain applications, such as the production of acetylene black, polyacetylene (which led to the 2000 Nobel Prize in Chemistry), and Reppe chemicals like 1,4-butanediol, which are difficult to replace with olefins.[22–25]

Acetylene production processes generally show a higher carbon footprint per kilogram compared to ethylene production when both are derived from fossil carbon sources.[13,26–28] Developing more sustainable C_2H_2 production methods through new technologies, non-fossil carbon, and renewable energy offers a promising strategy to reduce its carbon footprint.[13] However, research on alternative C_2H_2 production has lagged for decades, with most studies only focusing on C_2H_4 production.[29–31] As the industry shifts towards sustainability, reducing the carbon footprint of C_2H_2 industry presents an opportunity for its revitalization. In this review, we provide a comprehensive overview of C_2H_2 production methods, sustainable initiatives, and the challenges and opportunities associated with this transition.

2. Production of acetylene

2.1. Summary of acetylene production routes and its applications

Acetylene can be produced from fossil feedstocks such as coal, natural gas, and oil using four primary technologies, namely CaC_2, PC, plasma, and TP (**Fig. 2**).[32] Among these, oil-based routes have never reached the commercial stage. Plasma technologies include thermal plasma, where all generated species reach thermal equilibrium, and non-thermal plasma, where energy is preferentially channeled to electrons.[33] Thermal plasma, known as the electric arc, has extremely high energy consumption (12.1 kWh kg^{-1} C_2H_2),[5,34] and was therefore replaced by partial combustion (PC).[5] Plasma technologies for converting natural gas and coal to C_2H_2 continues at the pilot scale in China.[13,35] Thermal pyrolysis (TP) of hydrocarbons was commercialized by the Wulff process, which involved preheating bricks to 1300 °C followed by feeding hydrocarbons for cracking to produce C_2H_2.[6] However, due to a slow heating rate and significant soot formation compared to PC, Wulff plants have been shut down.[9] Currently, coal-derived CaC_2 and natural gas-derived PC technologies dominate C_2H_2 production. Notably, C_2H_2 production is region-dependent, with China producing over 8 Mt y^{-1} primarily from the CaC_2 route (~90%) due to its large coal reserves, while other countries produce less than 3 Mt y^{-1}, mainly via PC.[9,13] Additionally, C_2H_2 can be produced as an impurity in the steam cracking of hydrocarbons to olefins, typically removed by catalytic selective hydrogenation or solvent extraction.[28]

The first commercial application of C_2H_2 after 1894 was for lighting, which was later replaced by electric lighting. However, its use in oxyacetylene welding and cutting, introduced in 1903, remains widespread due to the flame temperature of approximately 3150 °C, making it one of the cheapest and most accessible methods for metalworking.[22] The most significant application of C_2H_2 is as a feedstock in the chemical industry,[31] with key downstream products summarized in **Fig. 2**. Currently, major applications of C_2H_2 include the production of vinyl chloride, 1,4-butanediol, acetylene black, and vinyl acetate. Vinyl chloride has the highest production volume, reaching approximately 50 Mt y^{-1},[36] predominantly used to produce PVC

with a market size of USD 68 billion. While the C_2H_4-based PVC route dominates globally, accounting for about 65% of PVC output,[37] approximately 78% of PVC production in China uses the C_2H_2 route, with the remainder using the C_2H_4 route.[21] Acetylene black has the smallest market size of approximately USD 0.3 billion.[38] It is a high-purity and highly crystalline form of carbon black produced from the thermal decomposition of C_2H_2, makes it an excellent conductive additive for electrodes.[9,38] Therefore, the market for acetylene black is expected to grow with the global shift towards electric vehicles.[38]

Another critical application of C_2H_2 is the production of 1,4-butanediol *via* the classical Reppe process, which involves the reaction of acetylene with formaldehyde. This process still dominates, accounting for 40% of total production (2–2.5 Mt y^{-1}).[39,40] Despite the advantages of the said route, several complex and multi-step alternative routes not relying on C_2H_2 currently exist, with maleic anhydride or 1,4-diacetoxy-2-butene as the intermediate.[39] The production of 1,4-butanediol exemplifies the unique advantages of C_2H_2 as the feedstock in chemical synthesis. However, research on converting C_2H_2 to chemicals has not attracted sufficient attention since being largely replaced by olefins.

The industrial production of vinyl acetate, acetaldehyde, acetic acid, acrylic acid, and acrylonitrile from C_2H_2 has been largely replaced by routes using C_2H_4, C_3H_6, and methanol. Acetaldehyde and vinyl acetate are primarily produced from C_2H_4 globally, with annual productions of 1.5 and 6.2 Mt, respectively.[41,42] Notably, vinyl acetate production in China still primarily relies on the C_2H_2 route to date. Acrylic acid and acrylonitrile are produced from C_3H_6, with annual productions of 8.8 and 7.0 Mt, respectively.[43–45] Acrylic acid has the largest annual global market size among the replaced commodity chemicals listed in **Fig. 2**,[43] at approximately USD 23 billion, while acetaldehyde has the smallest, at around USD 1.9 billion. Acetic acid, previously produced *via* C_2H_2-derived acetaldehyde oxidation, is now mainly produced by methanol carbonylation, with a high production of 18 Mt y^{-1} and a global annual market size of about USD 15 billion.[46,47]

2.2. Indirect acetylene production route *via* calcium carbide

In the CaC_2 route, unlike other methods, C_2H_2 production occurs at room temperature without thermodynamic limitation. However, the CaC_2 production itself occurs at high temperatures through the reaction between carbon and lime. CaC_2, also known as calcium acetylide, is a salt of the C_2^{2-} anion. The CaC_2 method allows for an easy reaction with water at room temperature to release the C_2^{2-} anion as C_2H_2. The production process of C_2H_2 *via* the CaC_2 route involves two reaction steps:

$$3C + CaO \rightarrow CaC_2 + CO \qquad \Delta H = 465 \text{ kJ mol}^{-1}$$

$$CaC_2 + 2H_2O \rightarrow C_2H_2 + Ca(OH)_2 \qquad \Delta H = -127 \text{ kJ mol}^{-1}$$

The formation of CaC_2 is highly endothermic, while the conversion of CaC_2 to C_2H_2 is exothermic.[48] Due to its simplicity and high selectivity for C_2H_2, the CaC_2 route still plays a dominant role in C_2H_2 production. This C_2H_2 manufacturing process (**Fig. 3a**) has low equipment investment but several disadvantages:[49] (i) the CaC_2 formation reaction requires temperatures of 2000 to 2300 °C to overcome the high endothermicity; (ii) large amounts of coal are consumed, with one-third of carbon resources converted to CO; (iii) significant CO_2 emissions result from the decomposition of limestone ($CaCO_3$) and the combustion of byproduct CO to provide heat for the arc furnace; (iv) large amounts of solid slag waste, mostly $Ca(OH)_2$, are produced. Due to these disadvantages and the transition of C_2H_2 to olefins, most CaC_2 plants in Western countries have shut down. However, in China, where coal reserves are abundant and imported oil for olefin production is costly, the CaC_2 route remains significant. After rapid growth in the 21st century (**Fig. 3b**),[15] CaC_2 production reached approximately 28 Mt y^{-1}, accounting for more than 96% of the world's production.[50]

In the CaC_2 route (**Fig. 3a, 3c**), producing 1 kg of C_2H_2 consumes approximately 3.1 kg of 80% pure CaC_2 derived from 2 kg of coal, while generating 2.8 kg of $Ca(OH)_2$ slag waste and 3.6 kg of CO_2.[9,13] Electricity consumption is primarily associated with the CaC_2 production process, requiring about 3 to 3.5 kWh kg^{-1} CaC_2,[51] and a total electricity consumption of about 10.6 kWh kg^{-1} C_2H_2.[11] Several process improvements have been proposed over the years, such as reusing $Ca(OH)_2$ slag waste as building materials or for lime production via dehydration,[18,52,53] CO_2 capture by $Ca(OH)_2$ to form limestone through mineralization,[48] CO recovery,[9] the use of renewable electricity,[54] and the use of biochar to replace coal.[14,53,55,56] These methods could reduce slag waste, CO_2 emissions, and coal consumption to some extent (**Fig. 3c**). Nevertheless, high electricity consumption remains inevitable due to the endothermic nature of the CaC_2 formation reaction, so CaC_2 plants are strategically placed in regions with cheap electricity. An oxythermal method, which heats a mixture of CaO, coke, and O_2, can produce CaC_2 with a lower energy requirement.[48] However, significant energy consumption is still necessary, often supplied by the combustion of coke. Other metal carbides (*e.g.*, BaC_2 and Li_2C_2) can be produced at relatively lower reaction temperatures, but they are yet to be commercialized due to more complex reaction pathways.[49,57]

2.3. Direct commercial acetylene production routes

Direct conversion of CH_4 to C_2 hydrocarbons (C_2H_2, C_2H_4, C_2H_6) is highly endothermic, with the CH_4 to C_2H_2 reaction being the most endothermic, having a standard enthalpy of 376 kJ mol^{-1}.[32]

$$2CH_4 \rightarrow C_2H_2 + 3H_2 \qquad \Delta H = 376 \text{ kJ mol}^{-1}$$

$$2CH_4 \rightarrow C_2H_4 + 2H_2 \qquad \Delta H = 202 \text{ kJ mol}^{-1}$$

$$2CH_4 \rightarrow C_2H_6 + H_2 \qquad \Delta H = 65 \text{ kJ mol}^{-1}$$

For the CH_4 to C_2H_2 process, the theoretical limit for the energy requirement is 4.0 kWh kg^{-1} C_2H_2.[58,59] Thermodynamic data show that CH_4 is the most stable hydrocarbon at normal temperatures, while C_2H_2 is the least stable (**Fig. 4a**).[32] With increasing temperature, the free energy of C_2H_2 decreases, making its formation more favorable than other hydrocarbons above 1230 °C.[32,60] As seen in **Fig. 4b**, CH_4 conversion to C_2H_2 and C_2H_4 is thermodynamically limited at 600 °C, but these limitations can be alleviated at 1200 °C.[30] Although C_2H_2 production becomes more thermodynamically favorable at high temperatures, these conditions also promote C_2H_2 decomposition to carbon and hydrogen (**Fig. 4a**). Thus, achieving reasonable C_2H_2 yields from CH_4 necessitates high reaction temperatures and rapid quenching of the C_2H_2 stream.

Two primary commercial technologies for direct C_2H_2 production are the PC and thermal plasma. These technologies have explored various carbon resources, including crude oil, naphtha, and coal, but natural gas remains the only feedstock currently used industrially. The reactor concepts are illustrated in **Fig. 4c**. In the PC process, natural gas and O_2 are preheated separately (to 600 °C when CH_4 is used) before mixing in the diffuser and passing through the burner and reaction chamber.[5] Approximately 60% of the CH_4 combusts with O_2 in the burner to stabilize the flame at a temperature above 1500 °C, facilitating the conversion of the remaining CH_4.[32] The reaction gas exits the reactor within milliseconds, followed by rapid quenching with water or oil sprays. This process results in a C_2H_2 concentration of around 8 vol.% in the cracked gas.[32] Thermal plasma, in contrast, uses an electric arc for direct heating,[61] achieving temperatures up to 20,000 °C at the arc center, while the electrode walls remain around 600 °C.[5] A vortex chamber stabilizes the arc between the cathode and anode by injecting gas tangentially. **Fig. 4c** highlights the critical role of reactor and process design in the industrialization of C_2H_2 production *via* PC and thermal plasma. **Fig. 4d** compares these two technologies.[5,9,13,32] The thermal plasma method yields higher C_2H_2 concentration in the cracked gas, lower natural gas consumption, and reduced direct CO_2 emissions. Despite these advantages, thermal plasma has significantly higher electricity consumption (12.1 kWh kg^{-1} C_2H_2) and extensive soot formation.[5,62] Another type of thermal plasma plant, operated by DuPont between 1963 and 1968, showed a higher C_2H_2 yield and lower electricity consumption of 8.8 kWh kg^{-1} C_2H_2 using a rotating arc.[63] However, the equipment design was more complex, with decreased stability, and costly H_2 was co-fed as the plasma gas.[63] With the increasing availability of green electricity, thermal plasma has garnered more attention.[13,21,64]

2.4. Emerging direct acetylene synthesis routes and catalytic applications

Recent advancements have explored non-thermal plasma technologies, including electrode-based plasma (dielectric barrier discharge (DBD), corona, pulsed, and spark) and electrodeless microwave plasma for C_2H_2 synthesis from CH_4 (**Fig. 5a**).[33] Non-thermal plasmas channel energy preferentially to electrons, creating a much higher electron temperature than that of ions and neutrals. When the neutral temperature is close to room temperature, the plasma is termed "cold plasma" (*e.g.*, DBD, corona, and pulsed plasma), whereas microwave and spark plasma can heat neutrals up to several thousand degree Celsius, making thermal activation significant; these are known as "warm plasma".[33] Electrode-based plasmas require electrodes for gas ionization to form plasma condition, while microwave plasma induces ionization *via* microwave irradiation, avoiding soot deposition issues on electrodes that affect plasma stability.[5] Reactor designs for non-thermal plasma technologies vary widely (**Fig. 5a**), necessitating case-by-case considerations to optimize energy efficiency. For example, DBD uses a dielectric barrier to prevent arcing and ensure uniform plasma discharge, while corona reactors rely on a pointy electrode to generate a localized, non-uniform electric field for plasma formation (**Fig. 5a**). The thermal pyrolysis (TP) of CH_4 in reactor tubes, heated either by electricity or combustion gas, can produce C_2H_2. However, this method often results in low conversion and significant C_2H_2 decomposition due to slow feed heating compared to PC. Adding H_2 can shift the equilibrium to inhibit C_2H_2 decomposition, but it also reduces the CH_4 conversion rate, making this approach commercially unviable.[32] Advanced reactor designs, including membrane reactors (**Fig. 5b**) or millisecond reactors, have been explored to enhance CH_4 conversion by shifting the reaction equilibrium toward the product side or reducing soot formation due to shortened contact time.[65,66] These reactor developments underscore the importance of reactor design in non-thermal plasma and TP technologies.

The comparison of representative results from thermal plasma, (catalytic) non-thermal plasma, and catalytic TP processes is summarized in **Fig. 5c** (**Supplementary Table 1 and 2**). Overall performance, based on CH_4 conversion and C_2H_2 selectivity, indicates that thermal plasma outperforms non-thermal plasma, which, in turn, outperforms TP processes (**Fig. 5c**). For non-thermal plasma, cold plasma's main advantage is eliminating the need for quenching since C_2H_2 is produced at low temperatures. In contrast, warm plasma typically achieves higher CH_4 conversion due to higher neutral temperatures (**Supplementary Table 1**). Feedstocks typically include pure CH_4 or CH_4 diluted in inert gas, with some attempts using co-feeds such as O_2,[67,68] H_2,[69] and CO_2.[70] However, non-thermal plasmas require substantial energy input, often exceeding Hüls arc plasma processes (**Fig. 5d**). Moreover, promising non-thermal plasma results (**Supplementary Table 1**) mostly derive from microwave or pulsed

plasma, which typically operate at low pressures or high frequencies, requiring complex equipment design for durable operation.[33]

In addition to reactor design, catalysts also play an important role in non-thermal plasma, with DBD and microwave being the most used (**Supplementary Table 1**). For instance, in DBD plasma, adding the MFM-300(Fe) catalyst increased C_2H_2 selectivity from 17% to 38% at CH_4 conversion of <3%.[71] In contrast, in microwave plasma, catalysts appear to have limited effect, with microwave irradiation being essential for C_2H_2 production. For example, microwave irradiation increased C_2H_2 selectivity from 0% to 35% at 700 °C with around 10% CH_4 conversion over a Mo/ZSM-5@SiC catalyst.[72] However, similar C_2H_2 selectivity was obtained under microwave irradiation with (33.7%) and without (38.4%) a Ni powder catalyst at 31% CH_4 conversion.[73] It is essential to consider thermal activation in warm plasma, as the role of high neutral temperatures cannot be overlooked. While extensive literature stresses the significance of non-thermal plasma on C_2H_2 production, detailed energy requirements are sometimes not provided, an essential indicator to evaluate the relevant technology.

For TP, the oxidative coupling of methane (OCM) and non-oxidative coupling of methane (NCM) can produce C_2 species, including C_2H_2, C_2H_4, and C_2H_6.[30] The slower heat transfer in TP compared to PC limits its commercial viability.[32] Therefore, TP could be competitive only if reaction temperatures are lowered (<1100 °C) using catalysts. OCM is thermodynamically more favorable than NCM, requiring lower reaction temperatures (450–1000 °C), but it is challenging to avoid C_2H_2 conversion or overoxidation to CO_x in the present of O_2.[74] Additionally, at low temperatures, the reaction of CH_4 to C_2H_2 becomes less thermodynamically favorable compared to the formation of other hydrocarbons (**Fig. 4a, 4b**). As a result, C_2H_2 production from TP primarily comes from NCM rather than OCM (**Supplementary Table 2**). The highest C_2H_2 selectivity reported is 35% at *ca.* 14% CH_4 conversion at 975 °C on a Pt_1/CeO_2 (Pt_1: Pt single atom) catalyst,[75] outperforming pristine CeO_2, Pt_n/CeO_2 (Pt_n: Pt nanoparticles). However, recent studies have revealed that Pt_1 in Pt_1/CeO_2 tends to sinter under reaction conditions, leading to the proposal of a Ce-Pt alloy or Pt_n as the potential active site.[76–78] Metal carbides, Fe_1 and Fe_n catalysts (**Fig. 5e**) also show good promise,[79–81] though CH_4 conversion is typically below 30% at temperatures <1100 °C. A CH_4 conversion of 63% is obtained on Pt on α-Al_2O_3 monolith (**Fig. 5c**), but a significant temperature gradient in the monolith from 200 to 1500 °C is observed, which is already close to the temperature in PC.[82] Additionally, coke deposition and soot formation are significant in catalytic NCM, though millisecond reactor designs can partially address this issue.[66]. Up to date, achieving high CH_4 conversion and high C_2H_2 selectivity simultaneously *via* catalytic TP remains challenging. C_2 products, especially C_2H_4 (**Fig. 5e**, **Supplementary Table 2**), are always present in large amounts, significantly affecting C_2H_2 selectivity. The presence of byproducts such as C_3-C_5 hydrocarbons, aromatics, and coke/soot further limits C_2H_2

formation.[80,83] Therefore, converting CH_4 to a broader range of products, such as C_2 or even C_{2+} hydrocarbons, should be a more practical route for future CH_4 utilization *via* catalytic TP. In contrast, cold and warm plasma routes can overcome thermodynamic limitations and achieve a higher yield of C_2H_2, though they usually require significantly higher energy input. Overall, advancing these emerging acetylene synthesis methods will require a focus on reactor design, catalyst development, and deeper understanding of reaction mechanisms.

3. The drive for acetylene's revival

The primary application of acetylene is in the production of vinyl chloride monomer (VCM) and subsequently PVC, a process that accounts for approximately 35% of global PVC production (**Fig. 2**). A production cost analysis (**Fig. 6a**) based on current data (refer to the **Supplementary Notes** for a detailed methodology) shows that the coal-based acetylene to VCM (ATV) scenario has a total cost of 0.6 USD kg^{-1} of VCM, making it more economical than the fossil-based (naphtha) ethylene balanced process (EBP) showing a total cost of 1.1 USD kg^{-1}. However, the climate change impacts (**Fig. 6b**) associated with ATV are notably higher than those of the EBP route (5.7 *vs*. 2.4 kg CO_2-eq kg^{-1}, assuming the same feedstock alternatives, respectively), primarily due to the more energy-intensive production of C_2H_2. This economic and environmental trade-off has led to the coexistence of both ATV and EBP routes in the market.

As the chemical industry moves towards sustainability, both ATV and EBP routes must explore more environmentally friendly alternatives, particularly the ATV route. By utilizing biochar-based C_2H_2 and C_2H_4 derived from the dehydration of bioethanol, a green scenario could emerge. The latter is already being deployed in countries like Brazil, using sugarcane as a sustainable feedstock. Interestingly, the cost of the biochar-based ATV route only slightly increases to 0.8 USD kg^{-1} when considering poplar as biomass source, while the cost of bioethanol-based EBP process rises significantly to 2.3 USD kg^{-1} (**Fig. 6a**), considering corn stover as feedstock. Although the climate change impacts in the green scenario remain higher for the ATV compared to the EBP route (1.7 *vs*. 0.8 kg CO_2-eq kg^{-1}), both routes show a significant reduction in carbon footprint simply by replacing the feedstocks. Notably, compared to the fossil-based EBP route, the biochar-based ATV route exhibits lower costs and reduced climate change impacts. Therefore, developing green acetylene production methods could drive a transition in PVC production from EBP to ATV routes, potentially revitalizing the acetylene industry. This shift could further enhance acetylene's market share by expanding its use in producing 1,4-butanediol, acetylene black, polyacetylene, and vinyl acetate (**Fig. 2**). Additionally, past acetylene applications, such as the production of acetaldehyde, acetic acid, acrylic acid, and acrylonitrile (**Fig. 2**), which have been replaced by other methods, might also

see a revival. Furthermore, new applications in chemical production could emerge, further expanding acetylene's relevance.[75]

Supplementary Table 3 presents the mass and energy balance, **Supplementary Table 4** displays the life cycle inventories used for the environmental assessment, and **Supplementary Table 5** lists the prices of feedstock and utilities used to estimate production costs for two VCM production routes. A detailed techno-economic and environmental assessment of various acetylene production technologies is provided in Section 5.

4. Renewable carbon resources for acetylene synthesis

Given the performance of biochar-based C_2H_2 in the ATV route (**Fig. 6a, 6b**), it is essential to explore sustainable C_2H_2 production methods using various renewable resources (*e.g.*, biochar, biomethane, CO_2, plastic) and innovative production technologies to replace traditional fossil-based approaches (**Fig. 6c**). Additionally, integrating renewable electricity (*e.g.*, wind and solar) into these processes is crucial for further advancing sustainability. **Fig. 6d** illustrates potential combinations of renewable carbon resources and relevant technologies for C_2H_2 synthesis. Biochar, derived from biomass pyrolysis, has been proposed for C_2H_2 synthesis *via* the CaC_2 route with a lower carbon footprint. However, it typically exhibits lower carbon and energy efficiency compared to the coal-based CaC_2 route due to the lower energy density of biomass and the generation of volatile components during pyrolysis.[14,53,55] Direct TP of various biomass feedstocks to C_2H_2 has been widely studied but is limited by selectivity towards C_2H_2.[61] Recently, the coal-based plasma route regained attention in China as a greener alternative to the coal-based CaC_2 route.[21] Biomass as a substitute for coal in the plasma route has also been explored, such as lignin and hemicellulose cracking in a hydrogen plasma, further reducing the carbon footprint.[84]

The pyrolysis of polyolefins and polyethylene terephthalate (PET) *via* rotating arc plasma has proven feasible for C_2H_2 production, achieving full carbon conversion and C_2H_2 selectivity of 55% for PET and 80% for polyolefins.[85,86] However, the specific energy requirement is approximately 20 kWh kg^{-1} C_2H_2, even higher than the Hüls arc plasma process. Biogas, as a renewable carbon feedstock composed mainly of CH_4 and CO_2, could replace natural gas for C_2H_2 synthesis with a lower carbon footprint. For instance, the conversion of biogas-like mixtures of CH_4 and CO_2 into C_2 products can be achieved using DBD cold plasma, with C_2H_6 as the dominant product.[87] Moreover, the addition of a catalyst (Ni–K/Al_2O_3) and increasing the reaction temperature above 220 °C can enhance C_2H_2 selectivity from biogas.[88] Direct conversion of biogas *via* TP and PC is challenging due to the presence of CO_2, despite CH_4 being a common feedstock for C_2H_2 production. However, the abundant availability of CH_4 or

as a byproduct of CO_2 utilization in biogas presents an opportunity to utilize biomethane for C_2H_2 production using those technologies like plasma, TP, PC (**Fig. 6d**).[31]

CO_2 in biogas or directly from the air can be integrated into the C_2H_2 production process *via* CO_2-CO or CO_2-C_2H_2 cycles. A sustainable route for C_2H_2 production, coupled with CO_2 capture and conversion to CO (CO_2-CO cycle), based on the $BaCO_3$-BaC_2-$Ba(OH)_2$-$BaCO_3$ cycle, has been reported (**Fig. 6e**).[49] Unlike the CaC_2 route, where CaO is obtained from $CaCO_3$ decomposition leading to CO_2 emissions, the CO_2-CO cycle does not directly emit CO_2. Additionally, BaC_2 synthesis requires lower temperatures, and less energy input compared to CaC_2. Notably, the carbon from CO_2 and half of the carbon from coal or biochar converts to CO instead of C_2H_2 in the CO_2-CO cycle (**Fig. 6e**), lowering atom economy if CO is not effectively utilized. In contrast, the proposed CO_2-C_2H_2 concept cycle directly transforms CO_2 into C_2H_2 *via* a cyclic electrochemical-assisted strategy (**Fig. 6f**).[57] This cycle involves three steps: (i) LiOH electrolysis to produce metallic Li^0 with H_2O and O_2; (ii) the active Li^0 reacting with CO_2 to form Li_2C_2 with Li_2O or $LiCO_3$ as byproducts; and (iii) hydrolysis of Li_2C_2 to form C_2H_2 and regenerate LiOH. CO_2 is used as the sole carbon input, presenting a totally green route. However, the recycling of Li_2O or $LiCO_3$ byproducts was not addressed. The CO_2-mediated concept cycle for C_2H_2 production shows most promise for sustainability, but it is still in early stages. Given the current foundation of coal- and natural gas-derived technologies, incorporating biochar or biomethane as potential green alternatives into those mature technologies appears more practical at this stage.

5. Environmental and economic assessment of acetylene synthesis routes

Recently, a framework was proposed to guide the development of sustainable chemical processes by prioritizing the environmental, economic, and equity (EEE) pillars.[89] To simplify the model, this section evaluates the environmental impact and economic viability of various acetylene synthesis technologies through techno-economic and life cycle assessment (LCA) analyses. We compare these technologies using both fossil and renewable carbon resources, assessing their environmental impacts within a future decarbonized scenario. For detailed insights into our methodology, please refer to the **Supplementary Notes**.

We specifically compare the climate change impacts in **Fig. 7a** and production costs (excluding capital expenditure) in **Fig. 7b** associated with the production of one kilogram of C_2H_2 using three different processes: CaC_2, PC, and plasma (both thermal and non-thermal). **Supplementary Table 6** shows the mass and energy balance, **Supplementary Table 7** displays the life cycle inventories used for the environmental assessment, and **Supplementary Table 8** lists the prices of feedstock and utilities used to estimate production costs for all C_2H_2 production routes. We evaluated two scenarios for all production technologies: fossil-based and green. For the CaC_2 process, we used coke (fossil) and biochar

(green) as feedstock. For the technologies utilizing methane as a feedstock in the fossil scenario (PC and plasma routes), we replaced methane with biomethane in the green scenario. Furthermore, we conducted a prospective life cycle assessment (LCA) to evaluate the projected climate change impacts for the year 2050 under a decarbonized scenario.

The highest climate change impacts are observed in the CaC_2 production scenario (12.9 kg CO_2-eq kg^{-1}) due to its heavy reliance on coke and electricity to produce calcium carbide, in addition to substantial direct emissions. In contrast, the PC has the lowest emissions (7.8 kg CO_2-eq kg^{-1}), utilizing methane as the primary feedstock and resulting in much lower direct emissions compared to the CaC_2 scenario. Both plasma technologies have higher climate change impacts than the PC route, primarily due to the high electricity requirements for these pathways. However, emissions are significantly reduced across all green scenarios. Notably, the PC route displays a negative climate change impact of –6.1 kg CO_2-eq kg^{-1} due to the utilization of biomethane feedstock, which results in negative net contributions to the overall impacts (on a cradle-to-gate basis) due to the biogenic nature of the carbon.[90] Similarly, the impacts for the green CaC_2 route show a 75% decrease due to the shift from coke to carbon-negative biochar.[91] In contrast to the significant reductions observed in the CaC_2 and PC routes, both plasma technologies show only a 45% decrease under the green scenario due to their heavy reliance on grid electricity, which is yet to be decarbonized.

In a future decarbonized scenario projected for 2050, emissions are expected to significantly decrease, primarily due to the adoption of a decarbonized electricity mix.[92] The most optimistic scenario (1.5 °C – RCP 1.9) is highlighted in **Fig. 7a** (purple scatter).[93] The CaC_2 route demonstrates a 63% reduction compared to current impacts under the fossil scenario and a 283% reduction under the green scenario. Similar reductions are observed across all production routes. These results underscore the potential of a decarbonized grid to mitigate climate change impacts, even in fossil-based production methods, and especially in the green scenario, as biochar also uses decarbonized electricity compared to the current scenario.

Regarding production costs within the fossil scenarios, the CaC_2 route has the lowest cost (1.2 USD kg^{-1}), primarily driven by electricity. The PC route is more expensive (2.0 USD kg^{-1}), largely due to methane costs. Similarly, both plasma routes have similar costs, averaging 1.7 USD kg^{-1}, predominantly due to electricity. In the green scenario, the CaC_2 route remains the most economical, with a 32% increase in costs due to the slightly higher cost of biochar compared to coke. The green PC scenario is the most expensive route, with a 340% increase in costs, driven by the substantial cost of biomethane (five times more expensive than methane). Similarly, plasma routes show at least a 110% increase in costs.

In the short term, the focus should be on optimizing the existing CaC_2 production process by transitioning from coke to a blend of coke and biochar, ensuring stable supply chains of biochar to maintain cost-effectiveness. The lower costs of biochar and its excellent reduction in climate change impacts make the green CaC_2 route the most favorable option for producing C_2H_2. In the long term, transitioning to renewable sources and engaging in grid decarbonization initiatives for electricity will further reduce the climate change impacts of plasma-based C_2H_2 production at lower costs. Moreover, lowering energy consumption in plasma processes will be key in the future. Comprehensive analysis is essential for engaging with stakeholders and advocating for policies to steer the chemical industry towards sustainable and economically viable C_2H_2 production.

6. Perspective and outlook

Transitioning from a fossil-based to a sustainable chemical industry is a global trend that presents an opportunity to revitalize the acetylene industry. To achieve this, increased focus on both C_2H_2 production and its chemistry is essential. For production, we consider two main strategies: integrating mature technologies with renewable carbon sources and decarbonized electricity to lower the carbon footprint, and exploring emerging technologies to improve C_2H_2 yield and energy efficiency. Integrating renewable carbon feedstocks (biochar or biomethane) and decarbonized electricity with mature technologies can efficiently reduce the carbon footprint, but it also increases production costs. Replacing natural gas with more expensive biomethane is not suitable for the PC route, as it requires nearly three times more biomethane than plasma technologies. Consequently, biochar-based CaC_2 and biomethane-based plasma routes are expected to become more prominent, with their development dependent on the availability and cost of biochar and biomethane, for which competitive uses (e.g., biochar for carbon removal and biomethane to defossilize industrial heating) will emerge. Given the explosive potential of C_2H_2, on-site generation for applications is advantageous.[32] The CaC_2 route offers a significant advantage by eliminating the risk of explosion through the transport of solid CaC_2 instead of C_2H_2, positioning the biochar-based CaC_2 route favorably for the future.

Emerging technologies, such as non-thermal plasma and thermal pyrolysis (TP), face commercialization challenges that require advancements in reactor design, catalyst development, and a deeper understanding of reaction mechanisms. Non-thermal plasma technologies show performance close to commercial thermal plasma but typically require higher energy input and more complex equipment. While catalysts can favor C_2H_2 production, the catalytic and plasma-assisted mechanisms are still not well understood. Future research should aim to improve C_2H_2 yield, energy efficiency, and equipment lifespan, with the goal of reducing energy requirements to the theoretical limit of 4.0 kWh kg^{-1} C_2H_2. Cold plasma offers advantages by preventing C_2H_2 decomposition and eliminating costly cooling processes, while

warm plasma usually achieves higher CH_4 conversion. The future dominance of either plasma type remains uncertain. Unlike non-thermal plasma, C_2H_2 yield through TP remains below 10% (**Fig. 5c**) due to thermodynamic limitations at low temperatures. To improve activity, selectivity, and catalyst stability, advanced catalyst design (*e.g.*, single-atom Pt_1 or Fe_1) and reactor design (*e.g.*, membrane or millisecond reactors) were developed. Despite decades of effort, C_2H_2 selectivity remains below 40% due to the formation of other hydrocarbons (**Fig. 5b**). Therefore, catalytic TP may be more practical for selective CH_4 conversion to C_2 products or mixtures of C_2 and other hydrocarbons. Additionally, kinetic studies and advanced *operando* characterization will be valuable in elucidating the complex CH_4 reaction mechanisms likely involving coupled surface and gas-phase chemistry. Designing high-temperature reactors that operate up to 1100 °C for synchrotron-based spectroscopy studies, especially under low-pressure conditions, will present a significant challenge in the future.

Electrocatalytic and photocatalytic CH_4 coupling has been widely studied, but the dominant products are C_2H_6 or C_2H_4 instead of C_2H_2.[94–96] Understanding why C_2H_2 production is inhibited in these novel processes is critical. Plasma-assisted plastic upgrading to C_2H_2 is promising but currently requires significant energy input (20 kWh kg^{-1} C_2H_2), making it impractical without further development.[85,86] Transforming CO_2 and H_2O into C_2H_2 and O_2 *via* the CO_2-C_2H_2 concept cycle is an ideal but nascent approach.[57]

Revitalizing the C_2H_2 industry also requires a substantial market for C_2H_2 applications. As discussed in Section 3, the development of sustainable C_2H_2 production could prompt a shift in VCM production from ethylene to acetylene routes based on the environmental and economic perspectives. Moreover, the longstanding challenge in C_2H_2-derived VCM production regarding the use of mercury catalysts has been addressed because of the recent commercialization of an Au/C catalyst and the development of advanced single atom catalysts for C_2H_2 hydrochlorination.[97–99] As a result, the shift in VCM production routes could stimulate the revival of the C_2H_2 industry. Since C_2H_2 chemistry was only extensively studied by Reppe in the mid-20th century, this revival might also reawaken the previously overlooked research on converting C_2H_2 into bulk chemicals.

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Author contributions

Z.Z. and J.P.-R. conceived the review. Z.Z. and A.N. produced figures. A.N. and G.G.G. completed the life cycle assessment and techno-economic study. Z.Z. and A.N. co-wrote the manuscript. G.G.G. and J.P.-R. revised the manuscript. J.P.-R. supervised the project. All authors reviewed, edited and enriched each section of the review.

Competing interests

The authors declare no competing interests.

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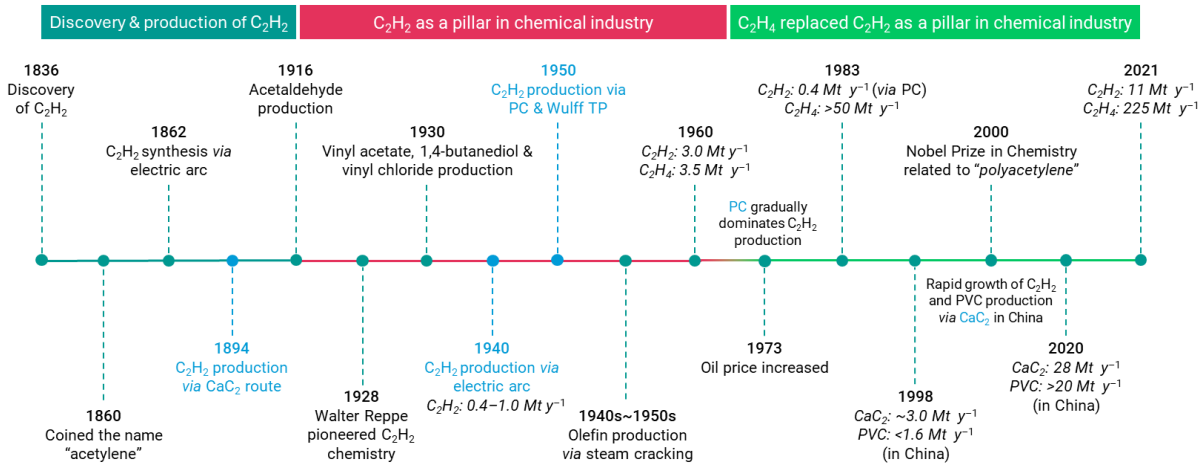


Fig. 1 | The acetylene timeline. The timeline illustrates the evolution of various industrial production technologies for C_2H_2 since its discovery, advancements in C_2H_2 chemistry, global production changes of relevant chemicals at key historical points, and the gradual market transition from C_2H_2 to C_2H_4.

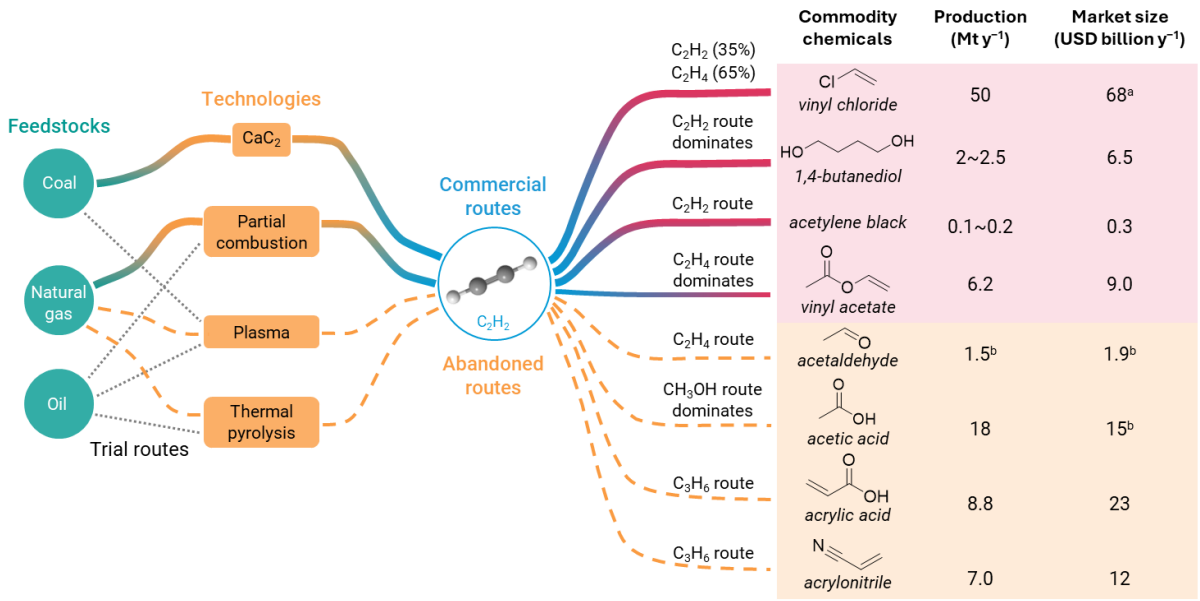


Fig. 2 | C_2H_2 production routes, past and current applications. The figure introduces various C_2H_2 production routes using fossil-derived feedstocks, including coal, natural gas, and oil, along with several technologies such as CaC_2, PC, plasma, and TP. The figure highlights major industrial applications of C_2H_2 and provides data on approximate global annual production (Mt y^{-1}) and market size (USD billion y^{-1}). It distinguishes between commercial routes (currently in use), abandoned routes (past commercial processes replaced by other methods), and trial routes (processes with research interest but never largely commercialized). [a]Since almost all VCM demand is for PVC production, this number reflects the global market size of PVC. [a,b]Data from Statista.

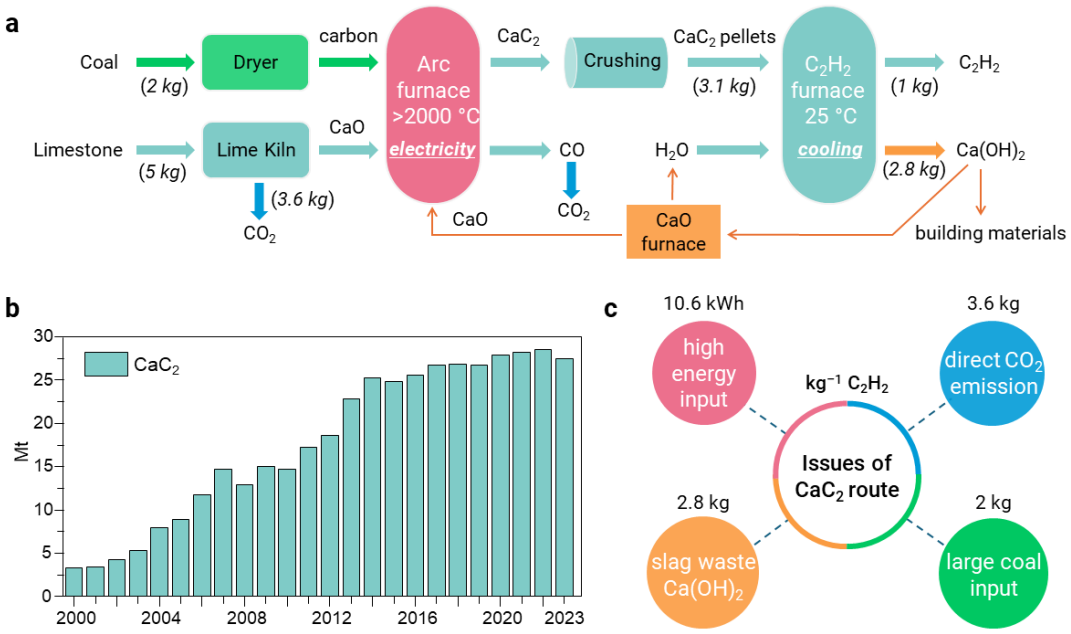
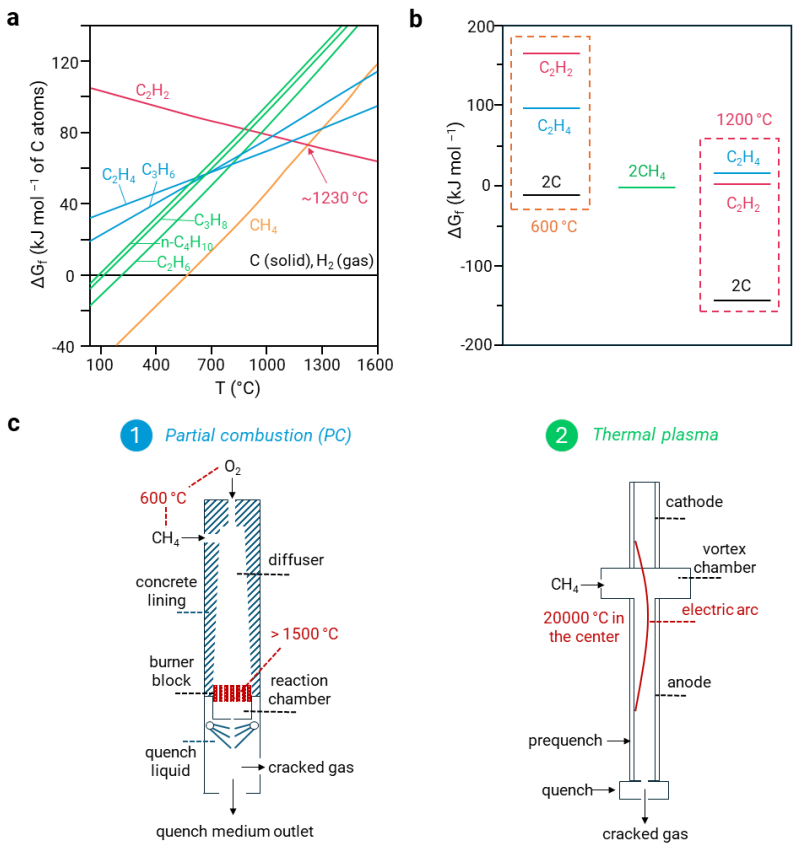


Fig. 3 | CaC_2-based route. a. The process for CaC_2-based C_2H_2 manufacturing. **b.** annual production of CaC_2 in million metric tons (Mt) from 2000 to 2023 in China. **c.** four major issues associated with the CaC_2 route, including high energy input, high direct CO_2 emissions, production of slag waste, and large coal input. Panel **b** adapted with permission from ref. 15, Elsevier.



d Comparison of two commercial technogies

C_2H_2 in cracked gas	8 vol%	11~15 vol%
energy input (kg^{-1} C_2H_2)	3.3 kWh	12.1 kWh
natural gas input (kg^{-1} C_2H_2)	5.7 Nm^3	2.0 Nm^3
CO_2 emission (kg^{-1} C_2H_2)	0.8 kg	~
soot formation (kg^{-1} C_2H_2)	50 mg	450 mg

Fig. 4 | Direct CH_4 to C_2H_2 production routes. a. Gibbs free energy of formation per carbon atom for several hydrocarbons from the elements as a function of temperature. **b.** Gibbs free energy of formation for C_2H_2, C_2H_4, and carbon from CH_4 conversion at 600 and 1200 °C. **c.** Reactor design for C_2H_2 production via PC and thermal plasma, and **d.** comparison of these two technologies. Panels adapted with permission from: **a.** ref. 32, Wiley; **b.** ref. 30, American Chemical Society; **c.** ref. 5, Royal Society of Chemistry.

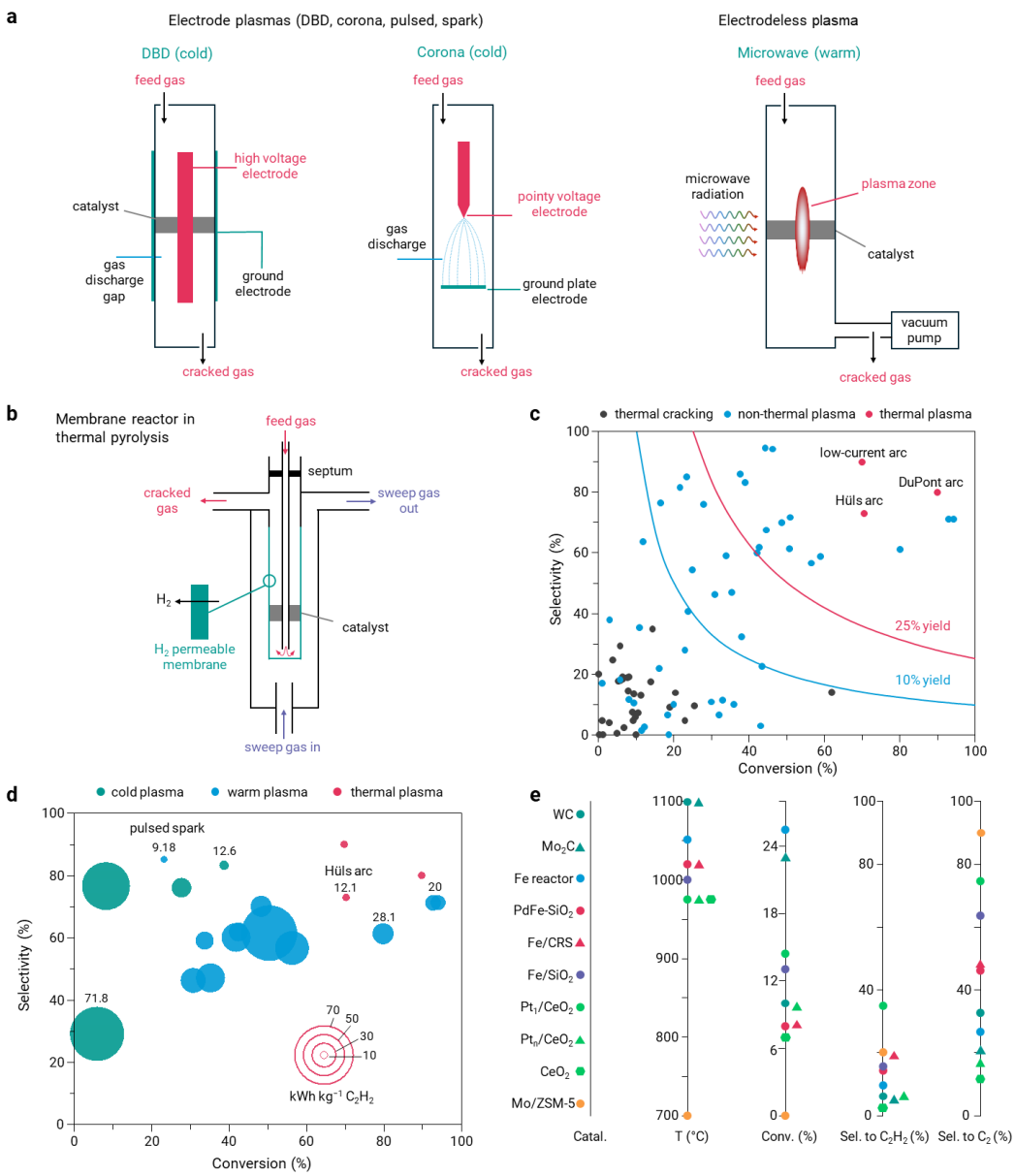


Fig. 5 | Emerging acetylene synthesis routes. a. Electrode- and electrodeless-based non-thermal plasma reactors.[5,69] **b.** Structure of membrane reactor in TP.[65] **c.** Conversion-selectivity trend in TP, non-thermal and thermal plasma. **d.** Conversion-selectivity trend and energy requirement (kWh kg^{-1} C_2H_2) for representative non-thermal (cold and warm) and thermal plasma routes. **e.** Representative catalyst type, reaction temperature, CH_4 conversion, and C_2H_2 and C_2 selectivity in non-oxidative CH_4 conversion. Panels adapted with permission from: **a.** ref. 5, Royal Society of Chemistry; **b.** ref. 65, Wiley.

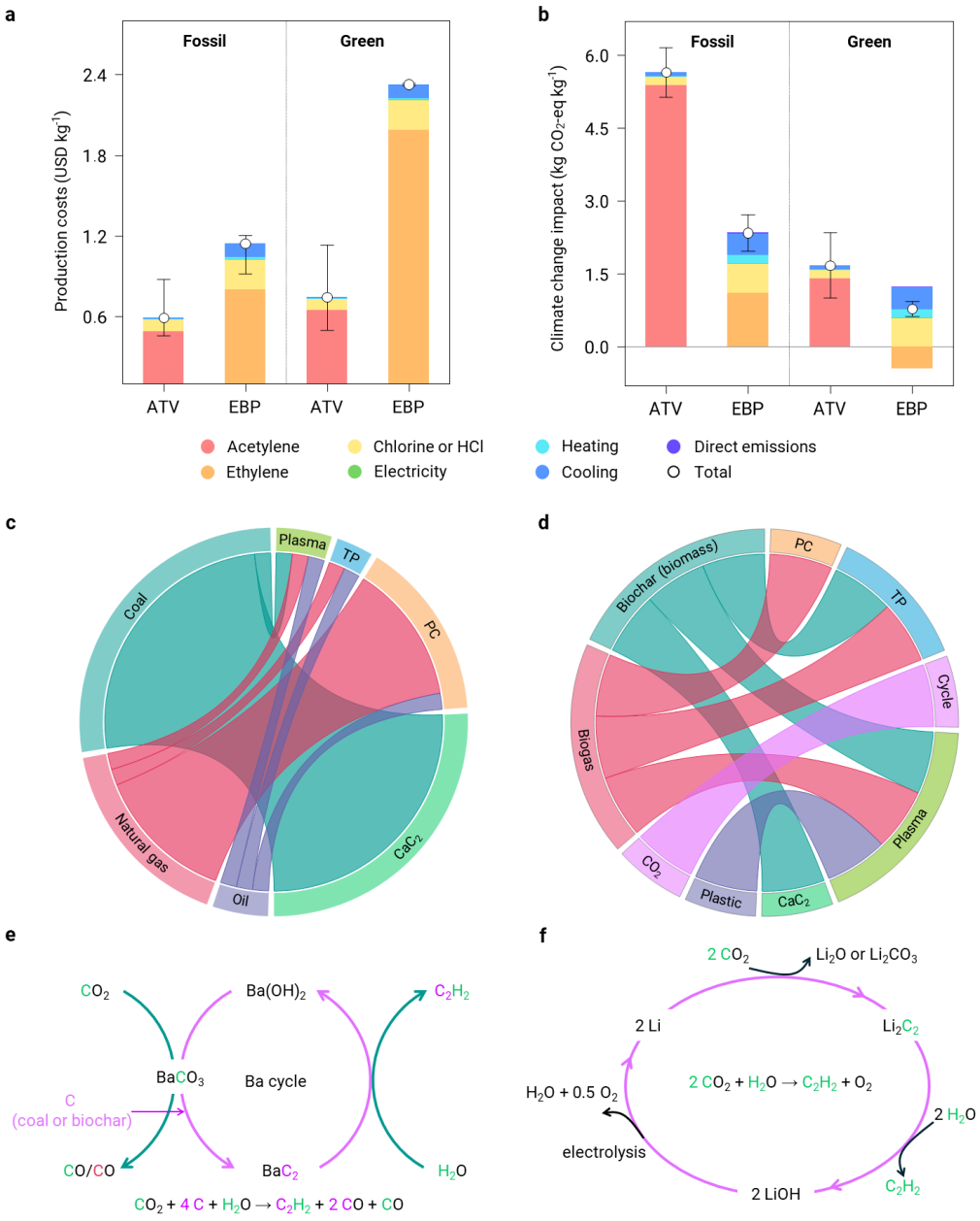


Fig. 6 | Potential green C_2H_2 synthesis routes. a. Production costs (only operational costs) and **b.** climate change impacts per kg of VCM *via* the acetylene-to-VCM (ATV) route and the ethylene-based balanced process (EBP). Results for both the fossil and green routes, depending on the feedstock source. Uncertainties (error bars) in both climate change impacts and production costs are displayed (see **Supplementary Notes** for details). **c.** Coal, natural gas, and oil-based technologies, including CaC_2, PC, TP, and plasma routes. The commercial routes, coal-based CaC_2 and natural gas-based PC, have a larger width due to their extensive use. **d.** biochar, biogas, CO_2, and plastic-based technologies, including CaC_2, plasma, cycle, TP, and PC. Renewable source-based routes are not yet commercialized, so their width remains uniform. CO_2-based concept cycles include **e.** BaC_2-based route coupling CO_2 capture and CO production (CO_2-CO cycle), and **f.** Li_2C_2-based electrochemical-assisted

three-step cycle from LiOH to Li, with Li reacting with CO_2 to form Li_2C_2, followed by reaction with H_2O to produce C_2H_2 (CO_2-C_2H_2 cycle). Panels adapted with permission from: **e.** ref. 49, Royal Society of Chemistry; **f.** ref. 57, Royal Society of Chemistry.

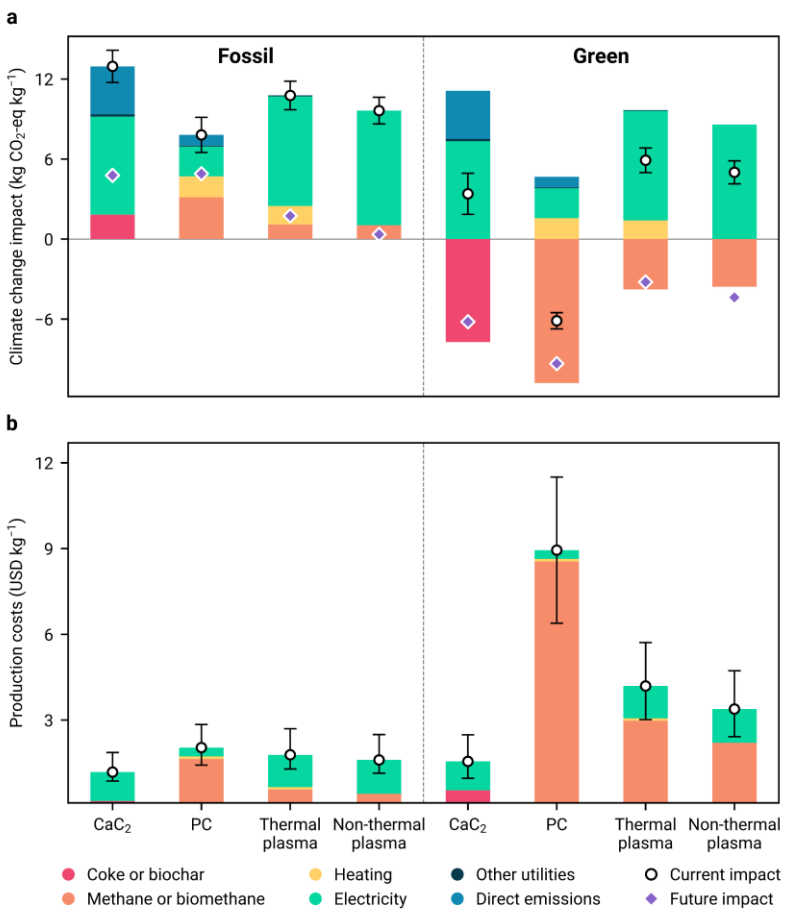


Fig. 7 | Environmental and economic comparison of C_2H_2 manufacture. a. Climate change impacts and **b.** production costs (only operational costs) per kg of C_2H_2 *via* the CaC_2 route and the PC or plasma routes. Biochar (from poplar), as an alternative to coke for the CaC_2 route, and biomethane, as an alternative to methane for the PC and plasma technologies, are evaluated for the green routes. Future projections (for 2050) of climate change impacts for all technologies are presented to illustrate the potential reduction in impacts under a decarbonized electricity mix scenario. Uncertainties (error bars) in both climate change impacts and production costs are displayed (see **Supplementary Notes** for details).